Vehicle Sales Trend Analysis: Market and Product Insights Over Time

MS Excel Project Created by Sayan Mukherjee

Introduction: This project analyzes BMW sales data to identify market and product trends over time, focusing on key variables such as fuel type, engine size, price, year, Units Sold and regional performance to support data-driven decision-making.

Objective: Identify Market and Product Trends Over Time.

Market Overview

Month

January | February | March | April | May | June
July | August | September | October | November | December

Fuel_Type

Diesel | Electric | Hybrid | Petrol

Total Units sold

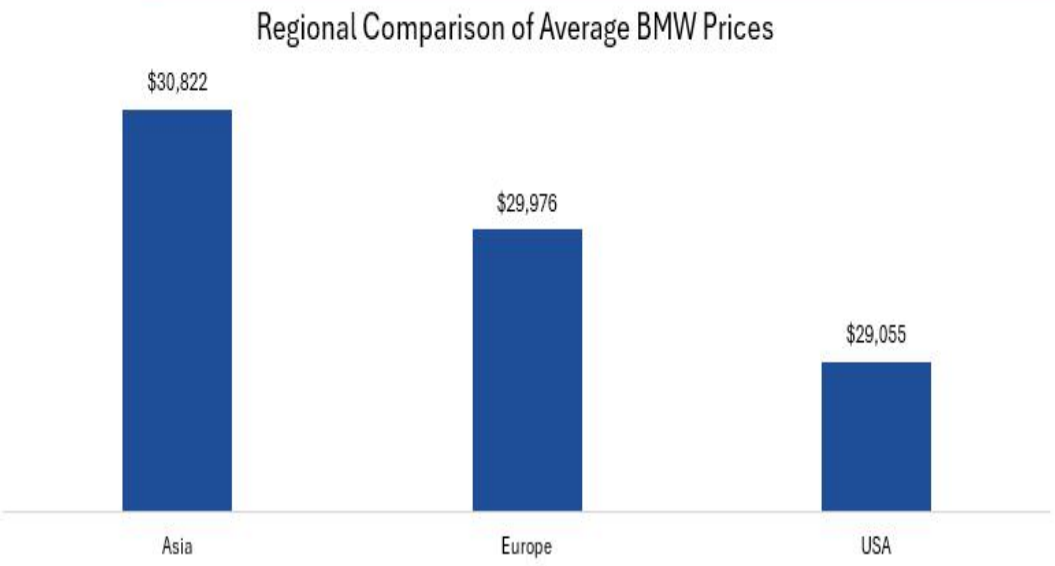
4,80,278

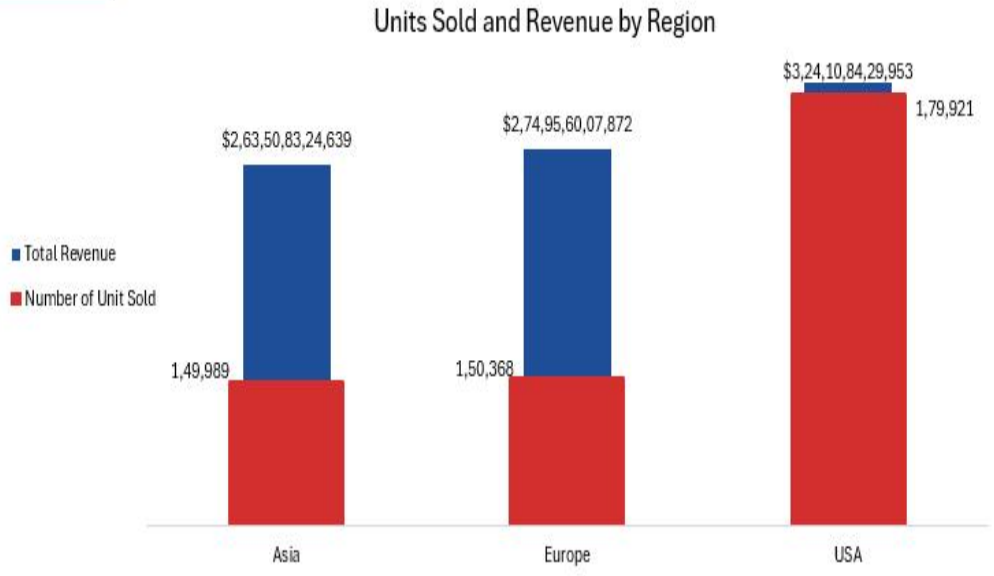
Total Revenue

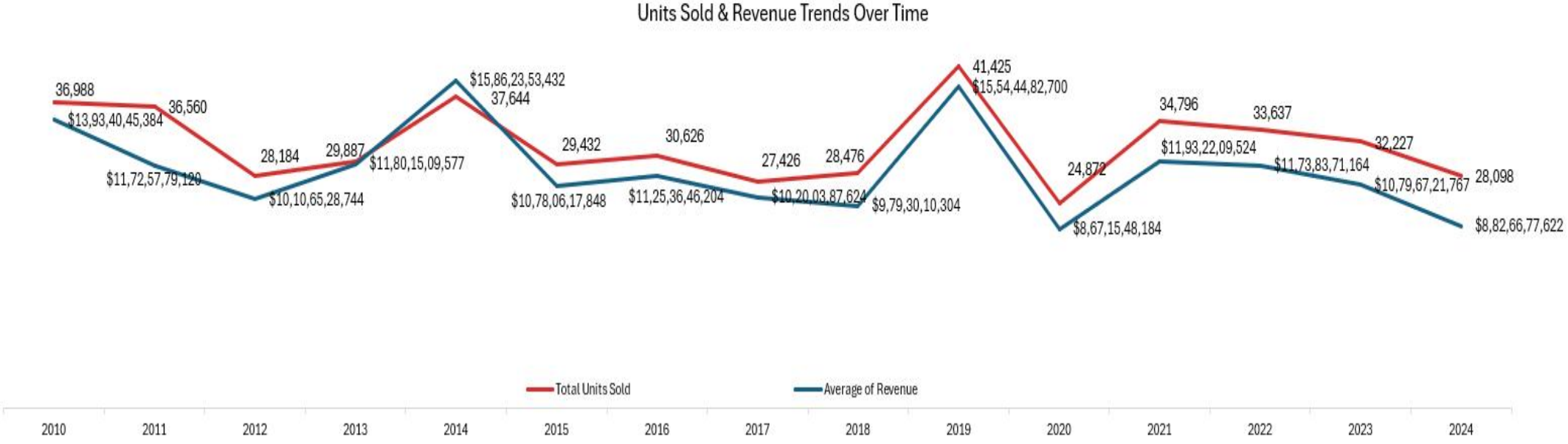
$25,87,16,05,69,844

Avg Price

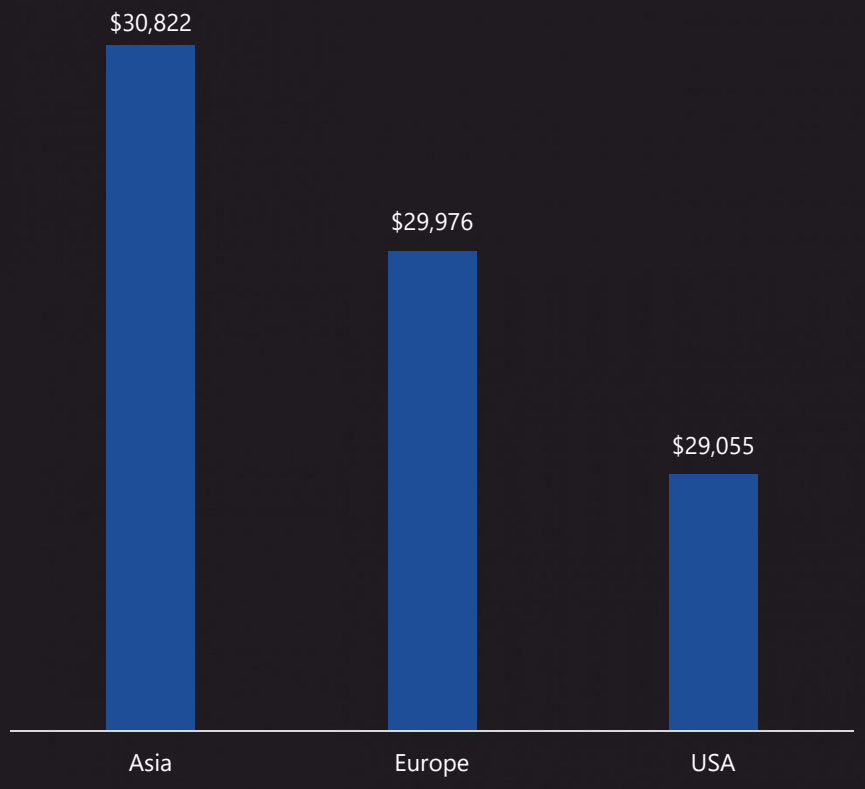
$29,927





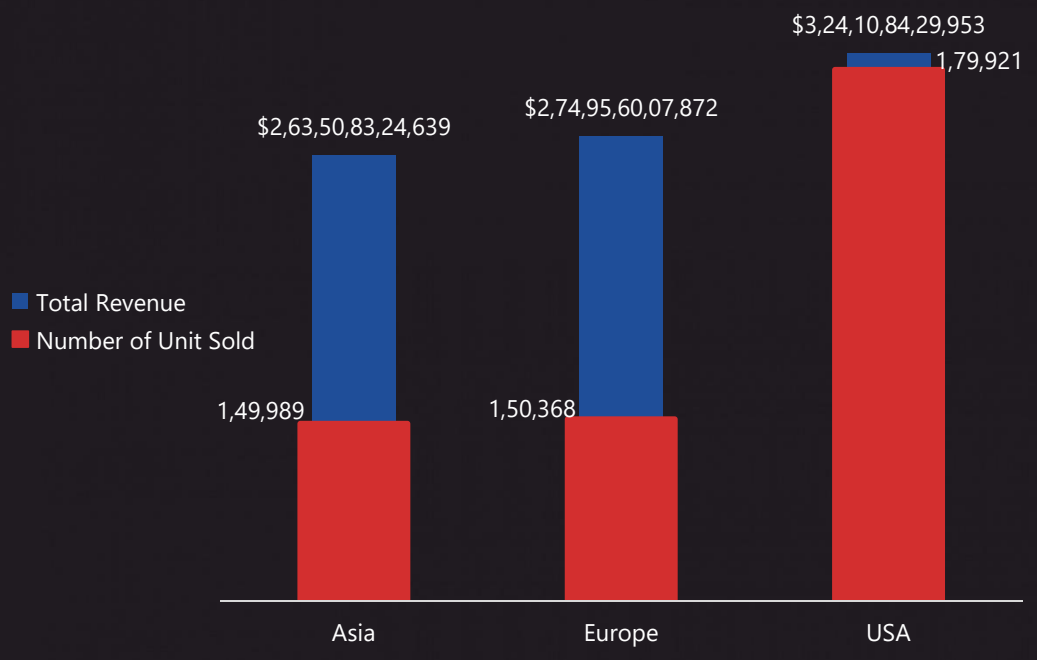


How does average price vary across regions?



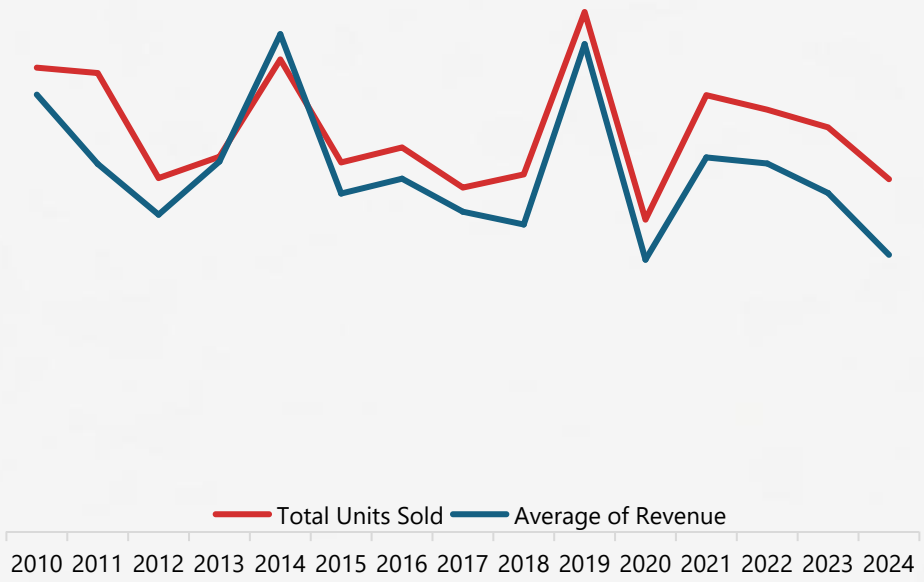
- **Insight:** Vehicle prices vary across regions due to differences in purchasing power and market demand. The United States has the highest average vehicle price (13.8% higher than Asia and 37.5% higher than Europe), indicating stronger demand for premium vehicles.

Which region generates the highest total revenue and Number of Unit Sold?



- **Insight:** The United States outperforms both Asia and Europe in terms of revenue and units sold, generating approximately 9.2% higher revenue than Asia and 3.6% more than Europe, while also achieving 9.5% higher unit sales than Asia and 7.8% more than Europe. However, the U.S. shows higher sales volume relative to revenue, indicating a lower average revenue per unit compared to Europe and Asia, where higher revenue per unit suggests stronger premium pricing. Overall, this highlights the U.S. as the most dominant market, driven by strong demand and high sales volume.

Are there increasing/decreasing trends in total units sold and avg revenue over the years?



- **Insight**: Both units sold and revenue peaked in 2019, dropped sharply in 2020 (likely due to external factors such as COVID-19), recovered in 2021, and then gradually declined from 2022 to 2024. However, between 2013 and 2015, revenue increased at a faster rate than units sold, suggesting a possible rise in average selling prices during that period.

Product, Technology & Performance Insights

Region: Asia | Europe | USA

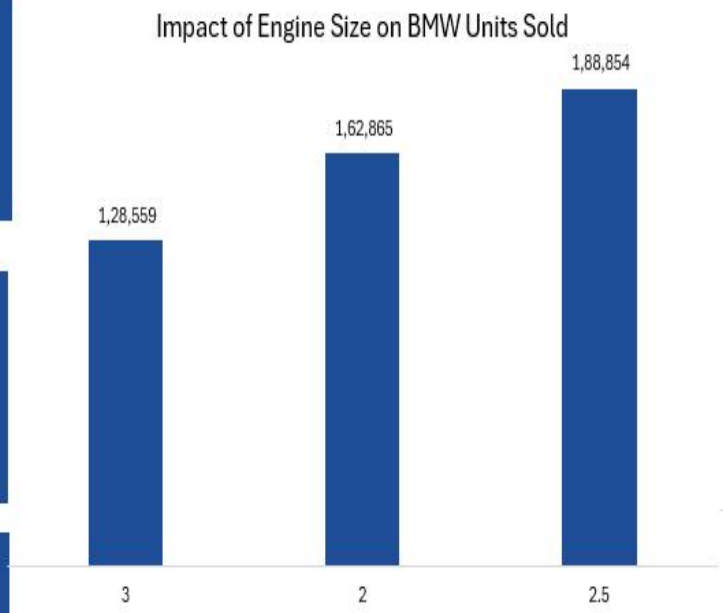
Month: January | February | March | April | May | June | July | August | September | October | November | December

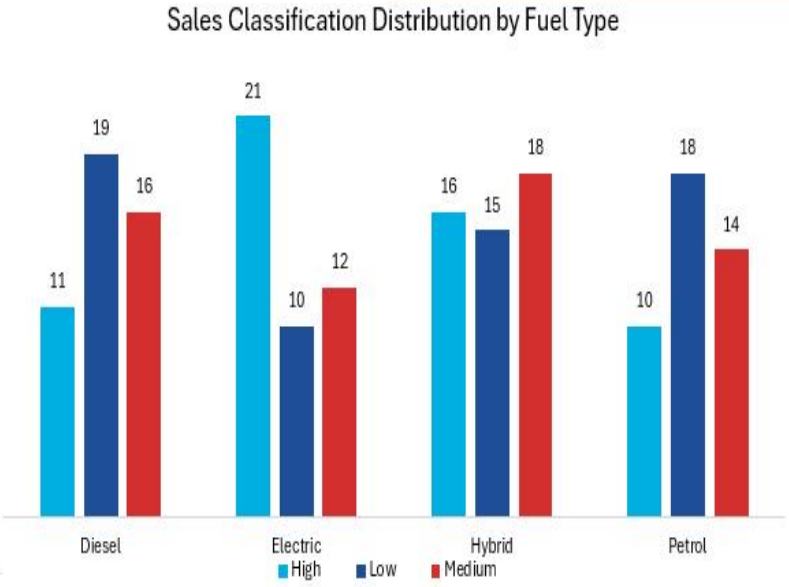
Top-Selling Model
3 Series

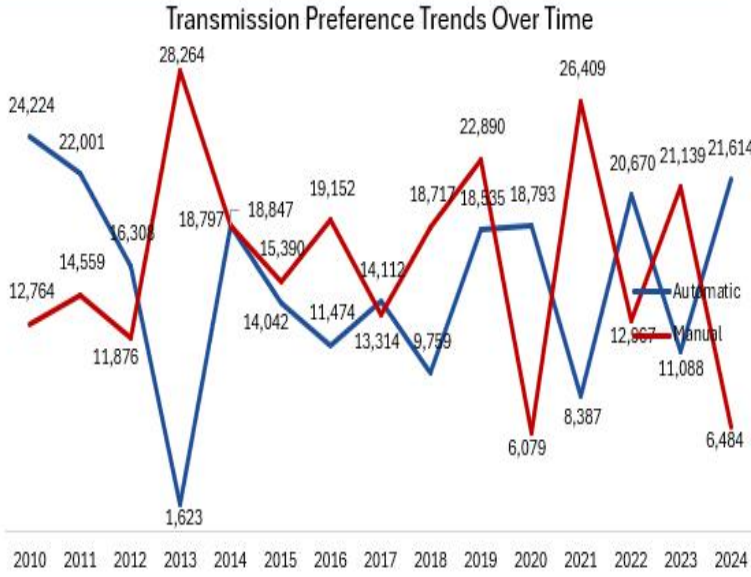
Low-Selling Model
X6

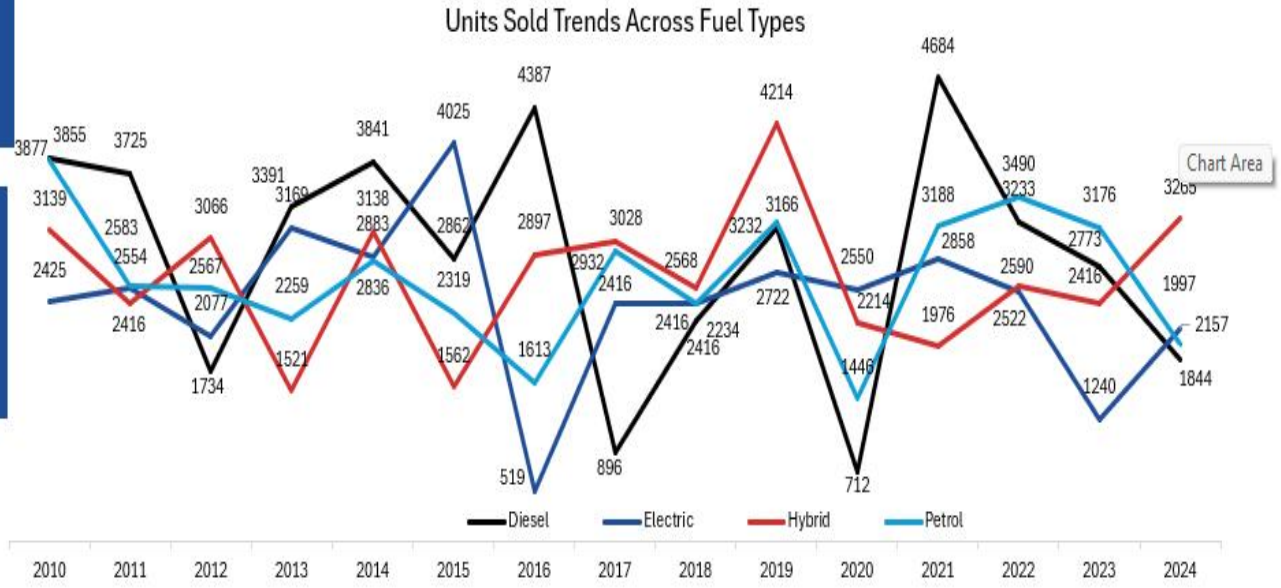
Average Engine Size
2.46

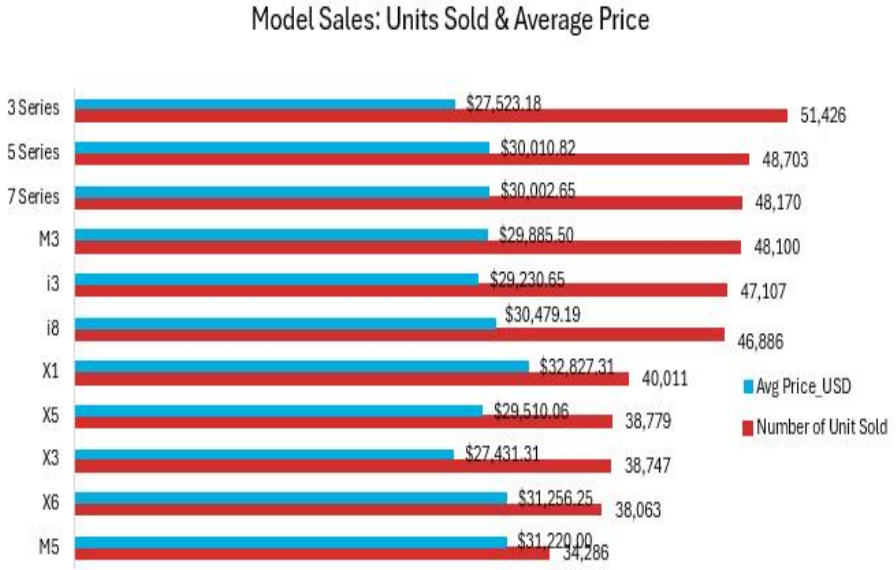
Average Mileage
70126.84



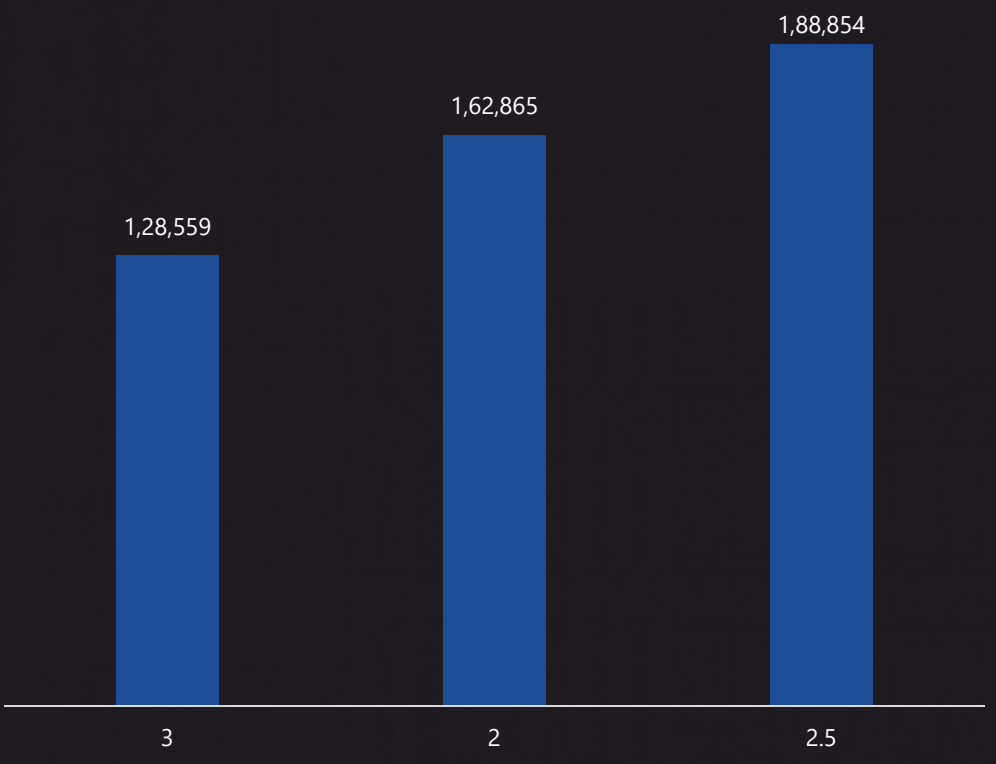






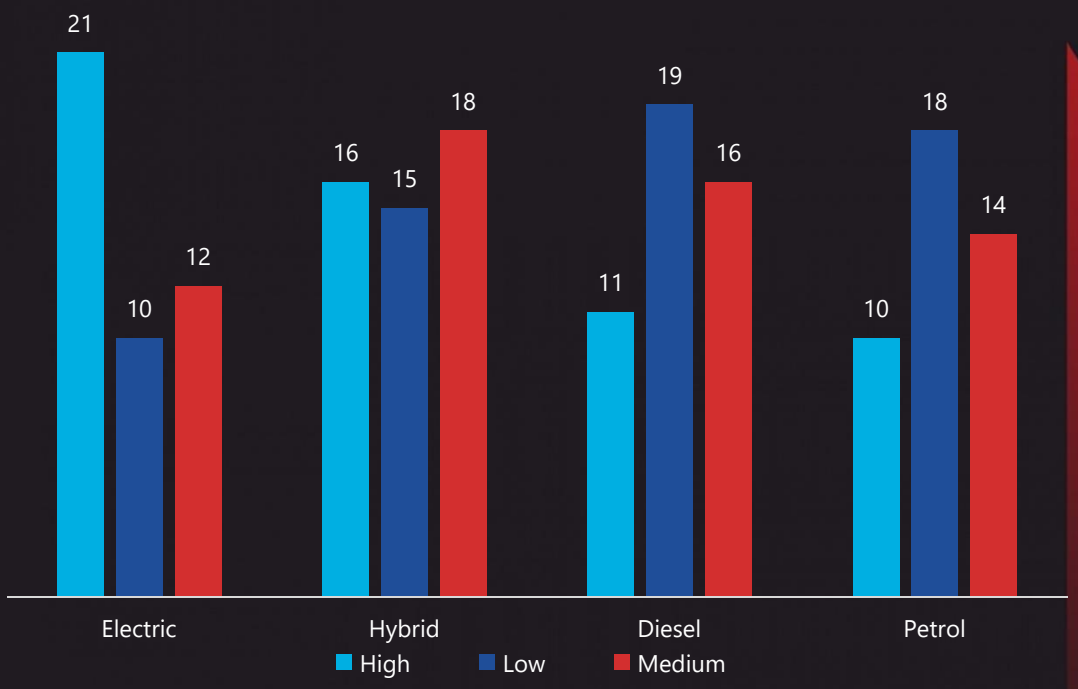


How does engine size impact units sold?



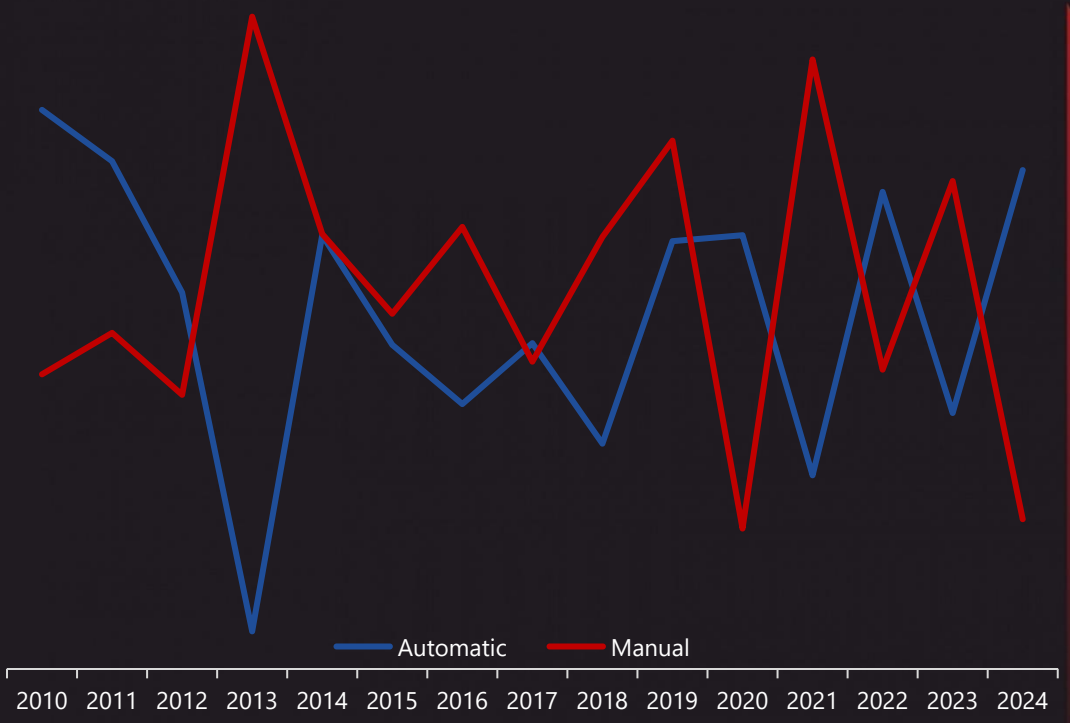
Insight: Mid-sized engines (2.5L) have the highest demand, followed by 3.0L, while 2.0L has the lowest sales. The 2.5L engine sold 59.5% more units than 2.0L and 16.8% more than 3.0L, suggesting that customers prefer mid-sized engines, likely because they offer a good balance between performance and fuel efficiency.

What characteristics define Sales Classification Fuel Type?



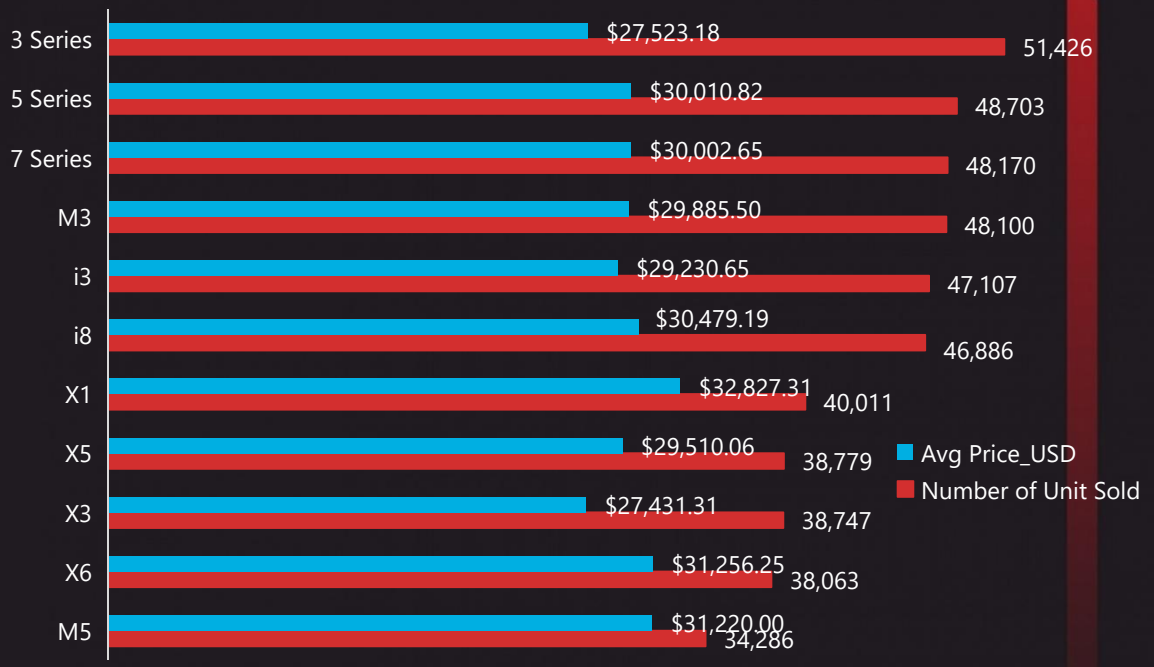
- **Insight:** Electric vehicles dominate the high-sales category, while hybrid vehicles lead the medium category. Diesel and petrol are more common in the low-sales category, showing a clear difference in sales patterns across fuel types.

How does transmission preference vary by year?



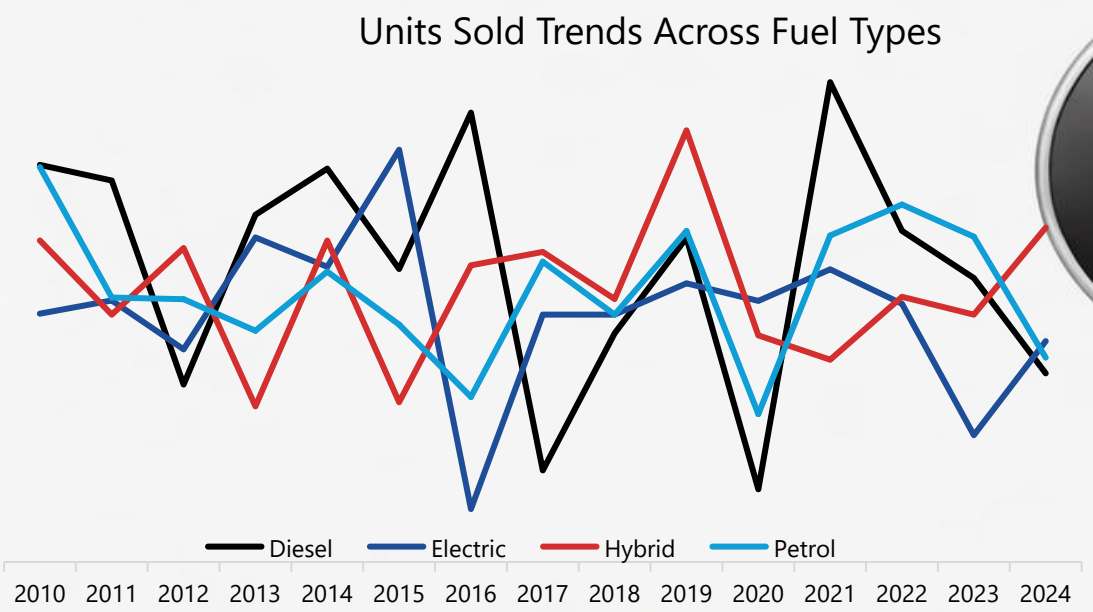
- **Insight:** Automatic transmissions dominated most years, but manual transmission sales spiked in 2013 and 2021, indicating temporary shifts in consumer preference or market conditions.

Which models perform best in terms of units sold and average price?



- **Insight:** Average price and units sold vary significantly across BMW models. The 3 Series achieves the highest sales despite its relatively lower price, while the X3 records lower sales even at a lower price point. Premium models like the M5 and X6 also show low sales volumes, indicating that pricing alone does not determine demand.

Are certain fuel types becoming more popular over time?



- **Insight:** Increasing but unstable electric and hybrid sales, alongside declining diesel trends, indicate a transition toward sustainable mobility, creating an opportunity to expand EV and hybrid offerings.

Recommendations:

- 1. Leverage the strong demand for mid-sized engines by increasing production capacity and introducing more model variations to maximize sales volume.
- 2. Focus on high-performing fuel types by aligning product offerings with customer demand, while gradually improving or repositioning underperforming fuel categories.
- 3. Optimize regional pricing strategies based on purchasing power and demand trends to maximize both revenue and sales volume.
- 4. Enhance underperforming models through product improvements, targeted promotions, and stronger value propositions.